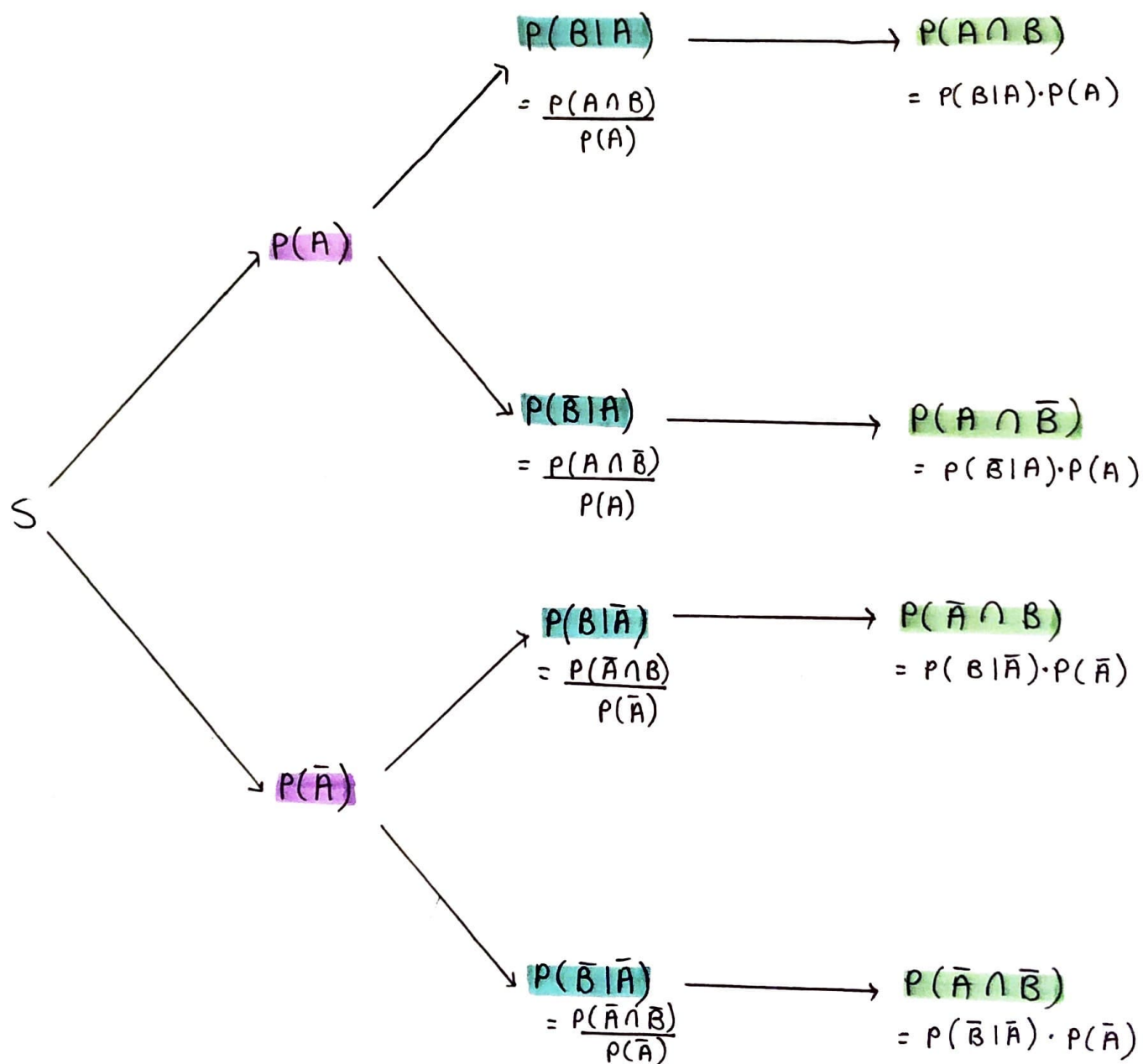
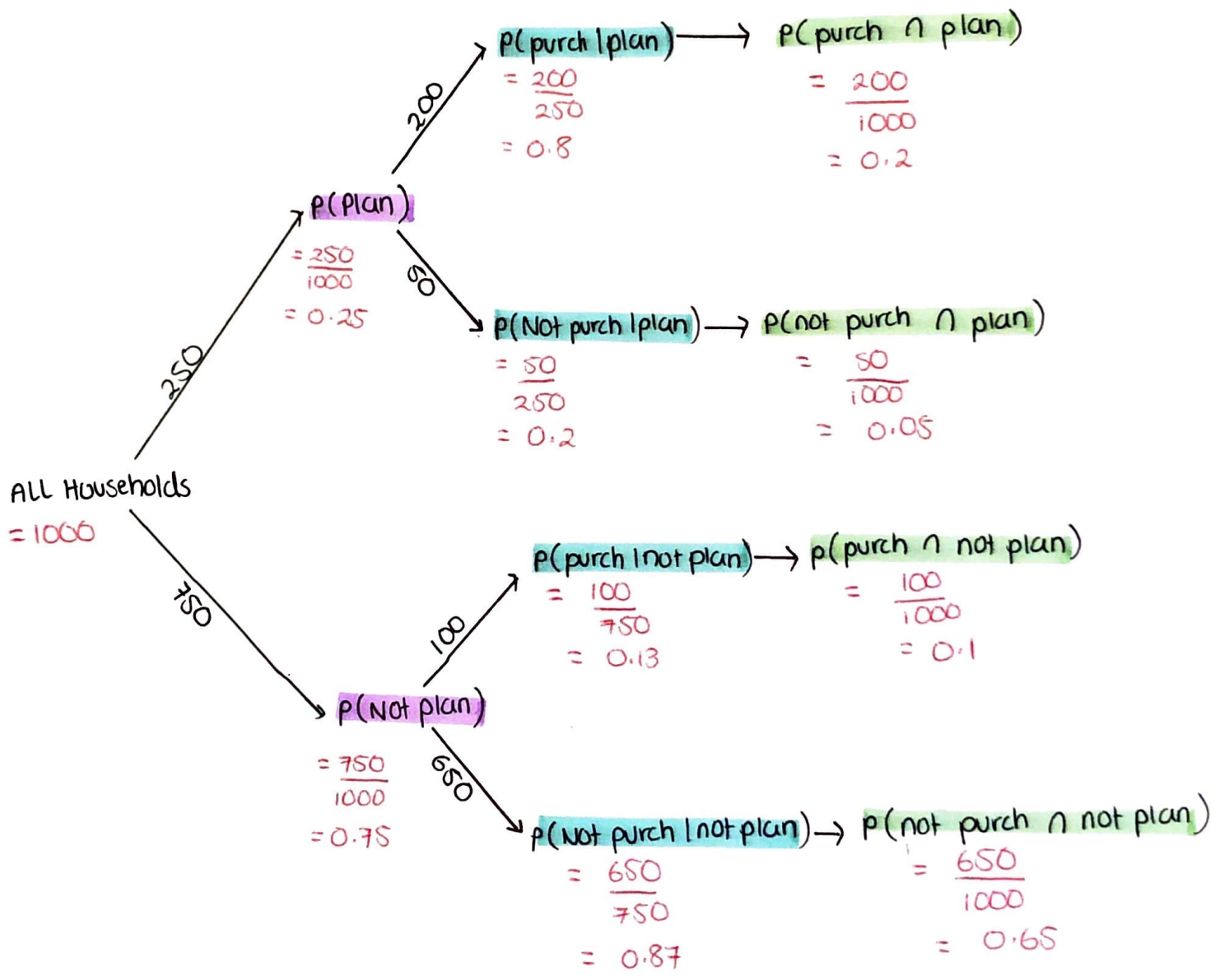


marginal, joint, and conditional probabilities from a decision tree



$$P(B) = P(A \cap B) + P(\bar{A} \cap B)$$

$$P(\bar{B}) = P(A \cap \bar{B}) + P(\bar{A} \cap \bar{B})$$



$$P(\text{purch}) = P(\text{purch} \cap \text{plan}) + P(\text{purch} \cap \text{not plan})$$

$$= \frac{200}{1000} + \frac{100}{1000} = \frac{300}{1000} = 0.3$$

$$P(\text{Not purch}) = P(\text{Not purch} \cap \text{plan}) + P(\text{Not purch} \cap \text{Not plan})$$

$$= \frac{50}{1000} + \frac{650}{1000} = \frac{700}{1000} = 0.7$$